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Key Points:

- The Xilinhot–Songliao Block (XSB) was located at ~19.3°N on the northern margin of the East Paleo-Tethys Ocean (EPTO) at 285 Ma
- The northern boundary of the EPTO tectonic domain extended to the southern margin of the XSB
- The Hegenshan Ocean between the XSB and Mongolia Block was the main ocean basin of the eastern Paleo-Asian Ocean at 285 Ma

Supporting Information:

Supporting Information may be found in the online version of this article.

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New Early Permian Paleomagnetic and Geochronological Data From the Xilinhot–Songliao Block and Their Implications for the Relationship Between the Paleo-Asian Ocean and the Paleo-Tethys Ocean

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Abstract The paleogeographic position of the Xilinhot–Songliao Block (XSB) is crucial to understand the spatiotemporal relationship between the Paleo-Asian Ocean (PAO) and East Paleo-Tethys Ocean (EPTO) tectonic domains. However, the lack of robust early Permian paleomagnetic data in the XSB has led to uncertainty in its paleogeographic position. Here, we report a well-dated (286.1–281.25 Ma) paleomagnetic pole (38.4°N/17.6°E, A_{95} = 4.0°) from andesites of the Dashizhai Formation in the XSB. We combined our paleomagnetic data with available geological and paleontological data to reconstruct the paleogeographic relationship between the PAO and EPTO. The XSB was located on the northern margin of the EPTO at ~285 Ma, suggesting that the northern boundary of the EPTO tectonic domain extended to the southern margin of the XSB, the Hegenshan Ocean (between the XSB and the Mongolia Block) corresponds to the eastern PAO, and the XSB separated the eastern PAO and EPTO at ~285 Ma.

Plain Language Summary The evolution of the Paleo-Asian Ocean (PAO) and the East Paleo-Tethys Ocean (EPTO) tectonic domains constructed the late Paleozoic paleogeographic pattern of Central and East Asia. Due to contradictions between the tectonic paleogeography and paleobiogeography of the Xilinhot–Songliao Block (XSB) during the early Permian, the spatiotemporal relationship between the two tectonic domains has been unclear. In this study, we present well-dated paleomagnetic constraints on the paleogeographic position of the XSB at ca. 285 Ma. We provide a new reconstruction for the Northeast Asian block, which suggests that the XSB was located between the eastern PAO and the EPTO during that time and was the intermediate unit that separated the two tectonic domains.

1. Introduction

The Central and East Asian microcontinents were important parts involved in Pangea formation (J. Zhou, Wilde, et al., 2018), and their paleogeographic evolution during the late Paleozoic was controlled by two tectonic domains (Figure 1a): the Paleo-Asian Ocean (PAO) and the East Paleo-Tethys Ocean (EPTO). Many tectonic reconstructions of the spatiotemporal relationship between these two tectonic domains have been proposed; however, it is generally thought that the intermediate units (Karakum–Tarim–North China; Figure 1a) that separated the two tectonic domains caused them to evolve independently (Domeier & Torsvik, 2014; Metcalfe, 2021; Scotese, 2021; Şengör et al., 2018; Wilhem et al., 2012; Xiao et al., 2015; Young et al., 2019; G. Zhao et al., 2018; Zuza & Yin, 2017). If this tectonic model is correct, then the two tectonic domains should have distinct endemic marine fauna during the late Paleozoic owing to the geographical isolation caused by the intervening tectonic units. However, early Permian paleobiogeographic data contradict this model. The tectonic reconstruction places the Xilinhot–Songliao Block (XSB) at mid–high latitudes on north of the PAO during the early Permian, in which case this block should have developed a cold-water Boreal fauna and bryozoan reefs. On the contrary, the warm-water Tethyan fauna (S. Z. Shen et al., 2006; Shi, 2006; C. Wang et al., 2014) and coral reefs (Kiessling & Flügel, 2002; Tian et al., 2011) developed on the southern margin of the XSB (Figure 1a). The current model cannot explain the lack of mixing between boreal and Tethyan fauna in the XSB during the early Permian

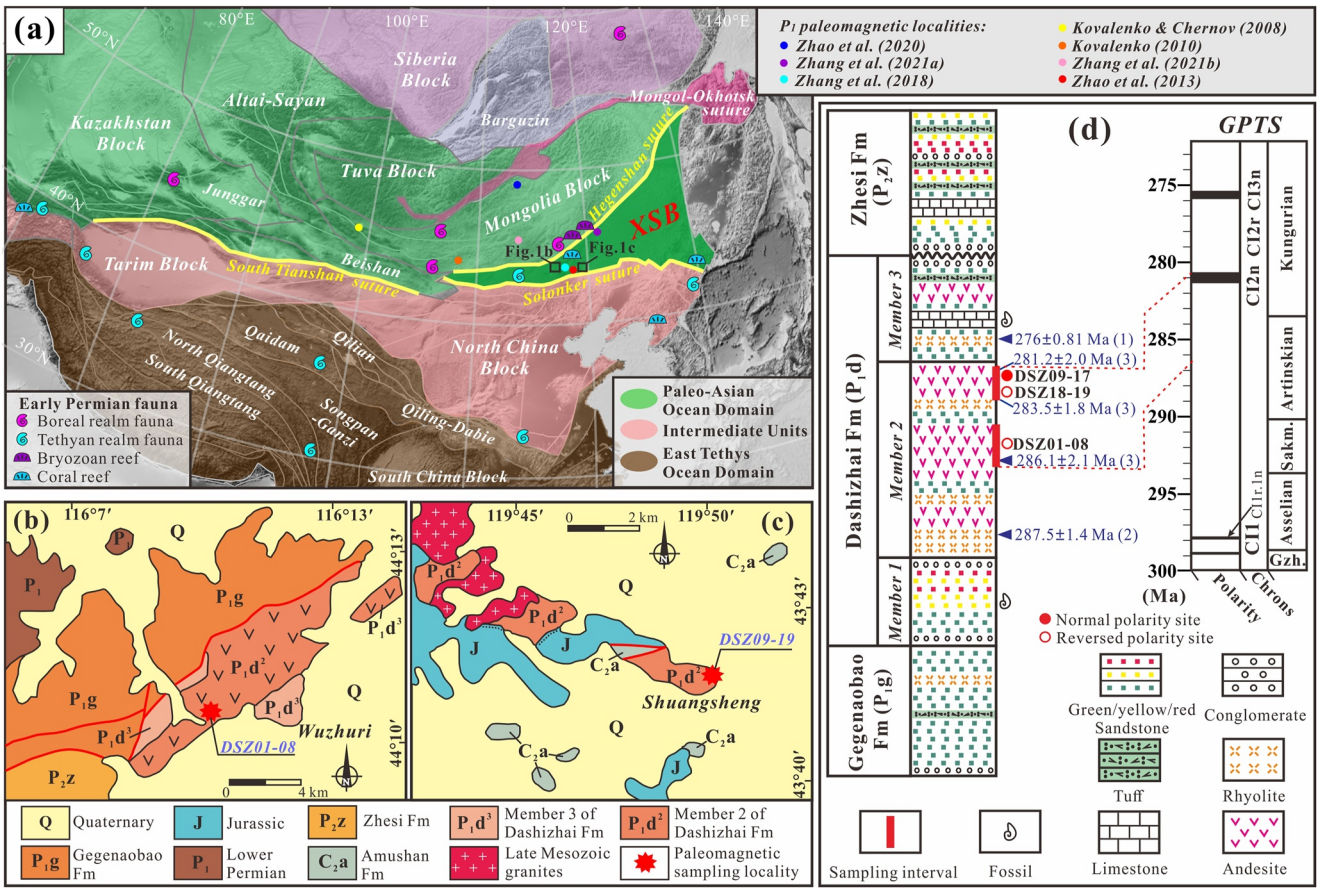


Figure 1. (a) Tectonic units of eastern Eurasia (modified from Zuza and Yin (2017); Huang et al. (2018); Ren et al. (2021)). Distributions of flora and fauna are based on C. Wang et al. (2014), X. Shen et al. (2019) and the Paleobiology Database (PBDB; https://paleobiodb.org/). The bryozoon and coral reefs are modified from Tian et al. (2011). (b, c) Geological map and (d) stratigraphic sequence in the Xilinhot and Shuangsheng sections in the eastern Xilinhot-Songliao Block (XSB; modified from BGMRIM, 1991). Geomagnetic Polarity Time Scale (GPTS) is modified from Hounslow and Balabanov (2018). Age references: (1) X. Zhang et al. (2008); (2) X. Zhang et al. (2016); (3) This study.

(Figure 1a; C. Wang et al., 2014). The main cause of this inconsistency is the uncertainty in the geographic position of the XSB during the early Permian.

We can use paleomagnetism to quantitatively estimate the paleogeographic position of tectonic plates; however, there are inconsistencies in the early Permian paleomagnetic data from the XSB (e.g., Li et al., 2012; Zhang, Huang, Meert, et al., 2021; D. Zhang et al., 2018; P. Zhao et al., 2013). Owing to the lack of precise age constraints, robust paleomagnetic tests (i.e., field, reversal, or paleosecular variation (PSV)), or well-defined tectonic units, estimates of the paleolatitude of the XSB range from ~40°N to ~5°N. As a result, three models have been proposed: (a) the XSB and the Mongolian Block (MOB) were located at mid–high latitudes, far from the North China Block (NCB; Domeier & Torsvik, 2014; Zhang, Huang, Meert, et al., 2021); (b) the XSB was connected to the MOB and NCB at low latitudes (Zhao et al., 2013, 2020); and (c) the XSB was an independent terrane near the paleoequator (Zhang, Huang, Zhao, et al., 2021; D. Zhang et al., 2018).

To ascertain the paleogeographic location of the XSB, we report new, well-constrained (ca. 283.7 Ma) paleomagnetic data from volcanic rocks of the Dashizhai Formation (Fm) in the southern XSB. We combine these data with paleobiogeographic data to reconstruct the paleogeography of the Northeast Asian blocks and reveal the spatiotemporal relationship between the PAO and EPTO domains.

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2. Geological Setting and Sampling

The XSB lies between the MOB and NCB, which is separated by the Hegenshan suture in the north and the Solonker suture in the south (Figure 1a; Y. Liu et al., 2017; S. Zhang et al., 2014). It is a relatively coherent tectonic unit with metamorphic basement (Xiao et al., 2003; S. Zhang et al., 2014). The XSB consists of two sub-blocks, the Xilinhot block in the west and the Songliao block in the east, which have been conjoined together since the Neoproterozoic (Y. Liu et al., 2017; J. Zhou, Wilde, et al., 2018). In this study, we conducted paleomagnetic and geochronological investigations in the Xilinhot and Balinzuoqi areas of the western part of the XSB (Figures 1b and 1c). The volcanic-clastic sequence of the lower Permian Dashizhai Fm is exposed extensively in this area, where it is disconformably overlain by sedimentary rocks of the middle Permian Zhesi Fm and conformably overlies the volcanic-clastic rocks of the lower Permian Gegenaobao Fm (Figure 1d; BGMRIM, 1991). The Dashizhai Fm is divided into three members: member 1 consists mainly of yellow, green, and red sandstones and conglomerates; member 2 consists of andesite and rhyolite interbedded with clastic rocks; and member 3 consists of andesites and limestones with intercalated tuff, sandstone, and rhyolite (Figure 1d; BGMRIM, 1991). The rhyolites at the bases of members 2 and 3 yield laser ablation–inductively coupled plasma–mass spectrometer (LA–ICP–MS) zircon U–Pb ages of 287.5 ± 1.4 Ma (X. Zhang et al., 2016) and 276 ± 0.81 Ma (X. Zhang et al., 2008), respectively. We collected a total of 120 oriented andesite paleomagnetic samples at 19 sites (DSZ01–DSZ19) over 2 sections (north Xilinhot and Shuangsheng) of the middle–upper parts of member 2. The attitudes of the andesitic lava were determined from crystal tuff and sandstone interbeds (Figure S1 in Supporting Information S1). Regional unconformities occur between the lower Permian Dashizhai Fm and the middle Permian Zhesi Fm (Figure 1d; BGMRIM, 1991), suggesting that the earliest deformation of the Dashizhai Fm took place during the late early–middle Permian. We obtained zircon U–Pb ages from three block samples of fresh andesite collected near paleomagnetic sampling sites DSZ08 (base of the north Xilinhot section), DSZ09 (top of the Shuangsheng section), and DSZ19 (base of the Shuangsheng section; Figure 1d).

3. Results

3.1. Zircon U-Pb Ages

Most of the zircon grains from three andesitic samples (DSZ1-1, DSZ1-2, and DSZ2) are euhedral to subhedral prisms, 90–160 μm by 40–60 μm, and have fine-scale oscillatory zonation (Figures S2a, S2d, and S2g in Supporting Information S1; detailed methods are presented in Text S1 of Supporting Information S1). Excluding ages with discordance >10%, sample DSZ1-1 (*Shuangsheng*) yielded a weighted mean $^{206}Pb/^{238}U$ age of 281.2 ± 2.0 Ma (MSWD = 0.106, n = 8; Figure S2c in Supporting Information S1), sample DSZ1-2 (Shuangsheng) yielded a weighted mean age of 283.5 ± 1.8 Ma (MSWD = 0.53, n = 12; Figure S2f in Supporting Information S1), and sample DSZ2 (north Xilinhot) yielded a weighted mean age of 286.1 ± 2.1 Ma (MSWD = 0.29, n = 10; Figure S2i in Supporting Information S1), indicating that the eruption age of the volcanic rocks in the paleomagnetic sampling sections was around 286.1–281.2 Ma.

3.2. Paleomagnetic Results

All 120 volcanic rock paleomagnetic specimens underwent stepwise demagnetization and yielded stable magnetic signals (Table S2 in Supporting Information S1; detailed methods in Text S2.1 of Supporting Information S1). Except for a few samples that yielded single components (e.g., Figure 2d), most yielded two well-defined components (Figures 2a–2c, 2e and 2f). In general, the in situ low-temperature components (removed below 250–300°C) are close to the direction of the recent local geomagnetic field (RGF; D = 0°, I = 62.6°, estimated using a geocentric axial dipole model; Figure S3 in Supporting Information S1) and represent viscous remanent magnetization by the RGF. Stable high-temperature components (HTCs) that decay toward the origin were isolated at 450–580°C in the samples from sites DSZ3–4, DSZ7, DSZ11–12, and DSZ15–19 and at 530–680°C in the samples from sites DSZ1–2, DSZ5–6, DSZ8–10, and DSZ13–14, thus carried mainly by magnetite and hematite, respectively (Text S2.2; Figure S4 in Supporting Information S1). The similarity of the remanence directions carried by magnetite and hematite (Figure S5 in Supporting Information S1) suggests that they were formed by high-temperature crystallization during initial cooling. The 19 sites yielded HTCs with a mean D_g of 165.1°, I_g of −32.8°, k of 8.3, and α_{95} of 12.4°, yielding a mean D_s of 125.4°, I_s of −34.2°, k of 65.3, and α_{95} of 4.2° after tilt correction (Figures 2g and 2h, and Table S2 in Supporting Information S1). The site-level virtual geomagnetic

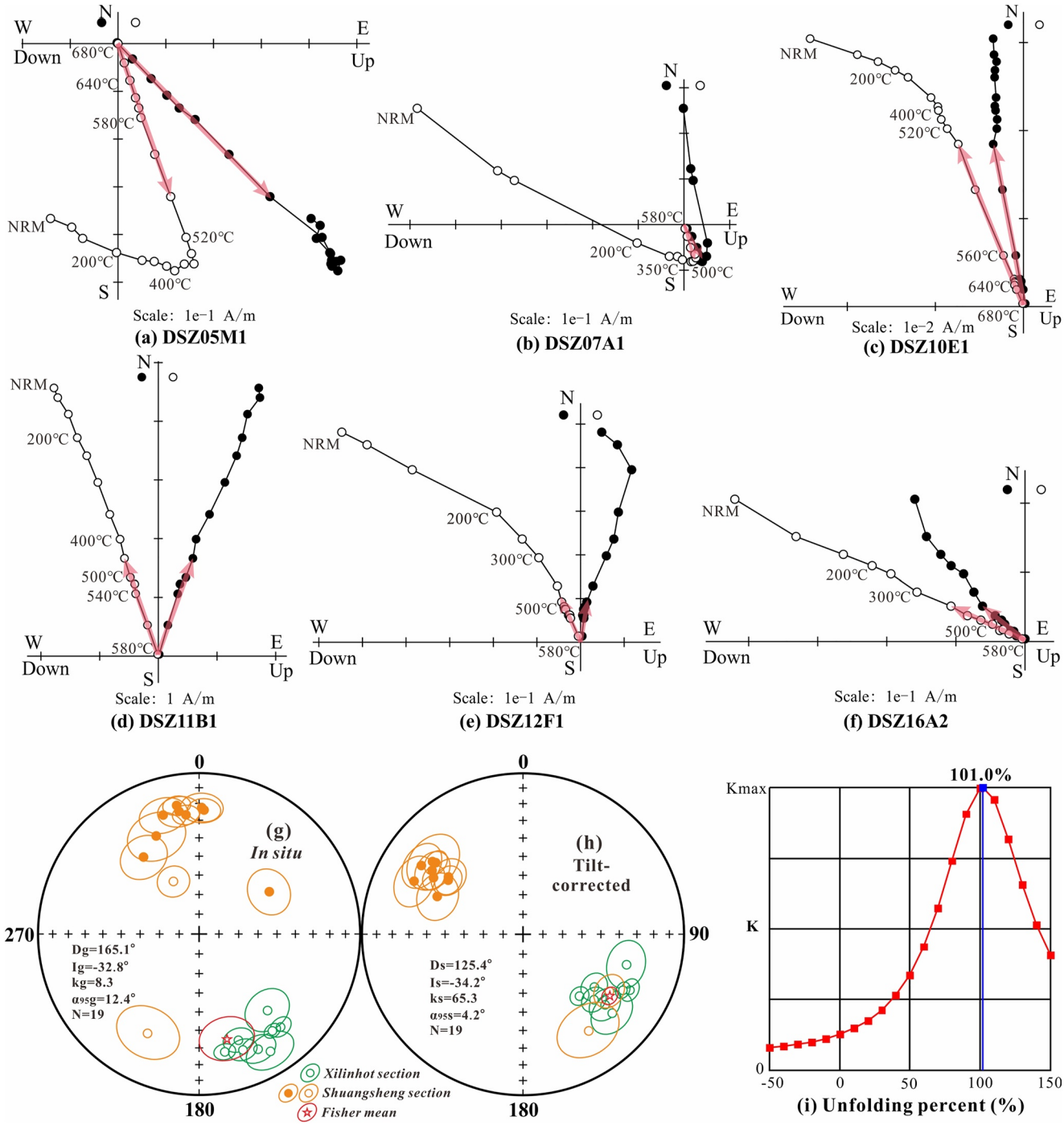


Figure 2. (a–f) Typical demagnetization characteristics of samples, in geographic coordinates. Solid/open symbols of the orthogonal plots represent the projections onto the horizontal/vertical plane. (g–h) Equal-area stereographic projections of the site-mean directions for the Dashizhai Fm. Lower (upper) hemisphere directions are represented by solid (open) symbols; stars indicate the mean directions with 95% confidence limits of 19 sites. (i) Step wise unfolding of the high-temperature components directions.

poles (VGPs) are Fisher distributed due to the positive results of the quantile–quantile plot test (Fisher et al., 1987; Mu = 0.808 < 1.207, Me = 0.943 < 1.094). The site-level HTCs passed a reversal test at the 95% confidence level (type B; McFadden & McElhinny, 1990) and a fold test at the 99% confidence level (McElhinny, 1964; McFadden, 1990). In a stepwise unfolding test, the precision parameter, *k*, reaches a maximum at 101% unfolding (Figure 2i; Watson & Enkin, 1993), which is within the critical error interval (90%–110% unfolding; Meert

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et al., 2020). These robust tests show that the HTC reflects primary magnetization. Our geochronological results indicate the remanent magnetizations were acquired at around 286.1–281.2 Ma during the reversed-polarity Kiaman Superchron (318–262 Ma; Opdyke & Channell, 1996). However, 9 of the 19 sites corresponding to our U–Pb age of 281.2 ± 2.0 Ma (DSZ1-1) yielded normal polarity directions (Figures 2g and 2h) and may record a short normal-polarity chron at 282–280 Ma (CI2; Figure 1d; Hounslow & Balabanov, 2018), which may explain the dual polarities observed during the Kiaman Superchron. In this paper, we assign a median age of 283.65 ± 2.45 Ma to our paleomagnetic samples from the Dashizhai Fm volcanic rocks.

Because quickly cooled volcanic rocks provide only a spot reading of geomagnetic field behavior (Meert et al., 2020), it is necessary to test whether the volcanic rocks of the Dashizhai Formation have averaged out the PSV. Some observations are as follows: (a) 19 paleomagnetic sites from at least 11 different lava flows span a long interval of time (286.1–281.2 Ma) and are interbedded with some layers of the volcanoclastic rocks in two sections; (b) the site-level VGPs are Fisher distributed; (c) the VGPs dispersion (S_B value) of 8.74 for 19 sites is in the range of predicted values of the Permian-Carboniferous Reversed Superchron "Model G" of de Oliveira et al. (2018); and (d) the value of A_{95}, obtained from the VGPs of 120 specimens, is 2.1, falling into an N-dependent A_{95} envelope with a 95% confidence interval (1.77, 4.02) proposed by Deenen et al. (2011, 2014). Thus, the PSV of the volcanic rocks have been sufficiently averaged out, and we define a mean pole at 38.4°N/17.6°E (A_{95} = 4.0°) by averaging all site-level VGPs (Table S2 in Supporting Information S1).

4. Discussion

4.1. New Paleogeographic Reconstruction of the Northeast Asian Blocks

The available late Carboniferous–Permian paleomagnetic data from the northeast Asian blocks are listed in Table S3 of Supporting Information S1 and shown in Figure 3a, and they are evaluated using the seven quality criteria of Meert et al. (2020). We emphasize that the published paleomagnetic poles must not show evidence of remagnetization that has been confirmed by a field or reversal test; therefore, we eliminated the data of Li et al. (2012) and Kovalenko and Chernov (2008). In addition, inclination flattening effects on paleomagnetic results from clastic rocks can complicate paleolatitudinal analyses. For the published paleomagnetic data, most researchers used a blanket flattening factor (f = 0.6) to correct the inclination of the clastic rocks (e.g., Torsvik et al., 2012; Van der Voo et al., 2015). This blanket correction is supported by the recent investigations of the lower Triassic red beds of the Liujiagou Formation (T. Zhou, Huang, et al., 2018) and the ~255 Ma volcanic rocks of the Qingfengshan Formation (Ren et al., 2020). In this study, the published data from clastic rocks are corrected with an optimal flattening factor, f, of 0.6 (Table S1 in Supporting Information S1).

Our new ca. 283.7 Ma paleomagnetic data from member 2 of the Dashizhai Fm in the XSB show that this block was located at 19.3° ± 4.0°N at that time for the reference point (44°N, 118°E; Figure 3b; Table S3 in Supporting Information S1; the same hereafter), which is different from the paleolatitude from the same formation reported by D. Zhang et al. (2018), probably because their paleomagnetic data were obtained from member 3 (280–270 Ma) of the Dashizhai Fm. The new paleomagnetic reconstruction placed the XSB at low latitudes between the NCB and the Tarim Block (Figure 4a), which corresponds to the development of extensive tropical–subtropical warm-water coral reefs in the southern XSB (Tian et al., 2011). Although the XSB and NCB have similar paleolatitudes (~20°N; Figure 3b; Table S3 in Supporting Information S1), we do not connect the two blocks in our reconstruction (Figure 4a) because the flora on the two blocks belonged to different realms during the early Permian (Figure 4a; Sun, 1997), pelagic–hemipelagic sedimentary successions developed on the southern margin of the XSB at that time (Tian et al., 2016), and early Permian subduction-related magmatism occurred between the NCB and XSB (Eizenhöfer & Zhao, 2018; Y. Liu et al., 2017; Xiao et al., 2018). These observations indicate that both the southern XSB and northern NCB margins bordered a wide ocean during the early Permian.

P. Zhao et al. (2020) and Zhang, Huang, Zhao, et al. (2021) published early Permian paleomagnetic data that suggest the MOB was located at ~20.1°N and ~46.5°N, respectively, indicating two potential paths of paleolatitude evolution (Figure 3b). However, the paleomagnetic pole of P. Zhao et al. (2020) may not have fully averaged out the PSV. In addition, cold-water boreal fauna (rather than warm-water Tethyan fauna) were present on the southern margin of the MOB during the early Permian, and extensive temperate bryozoan reefs (rather than tropical–subtropical warm-water coral reefs) developed in the southern MOB (Figure 4a; Tian et al., 2011),

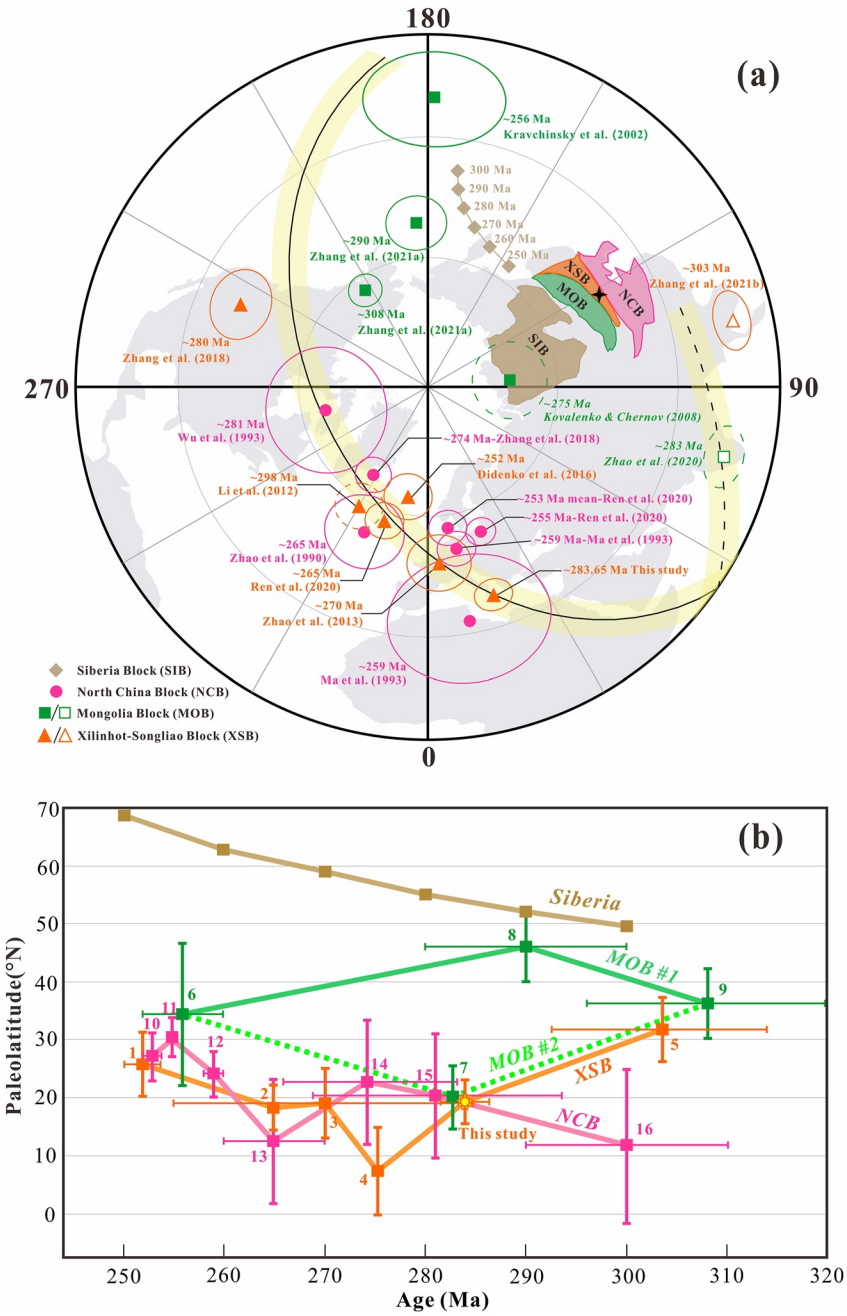


Figure 3. (a) Comparison of the Permian paleomagnetic pole in the equal-area projection. (b) The paleolatitude curves versus age. Data serial numbers see Table S3 in Supporting Information S1.

suggesting that the MOB was located at mid–high latitudes during the early Permian (Kiessling, 2001). We thus assume version 1 (MOB#1; Zhang, Huang, Zhao, et al., 2021; Figure 3b) is correct in the discussion below.

Our paleomagnetic data show that there was a large difference (~20°) in the latitudes of the XSB and MOB at ca. 285 Ma (Figure 3b), which suggests that large-scale lithospheric shortening has since occurred. Additional geological observations support this hypothesis. For example, the Hegenshan ophiolite, which represents oceanic crust that formed during the early Permian, was likely emplaced no later than the latest Permian (Miao et al., 2008). Early Permian subduction-related magmatism is recorded on the southern margin of the MOB (Eizenhöfer & Zhao, 2018; Song et al., 2015; Xiao et al., 2018), and marine deposition occurred across the Hegenshan Suture (e.g., the Xilinhot and Hegenshan regions) during the early Permian, as indicated by the fossil

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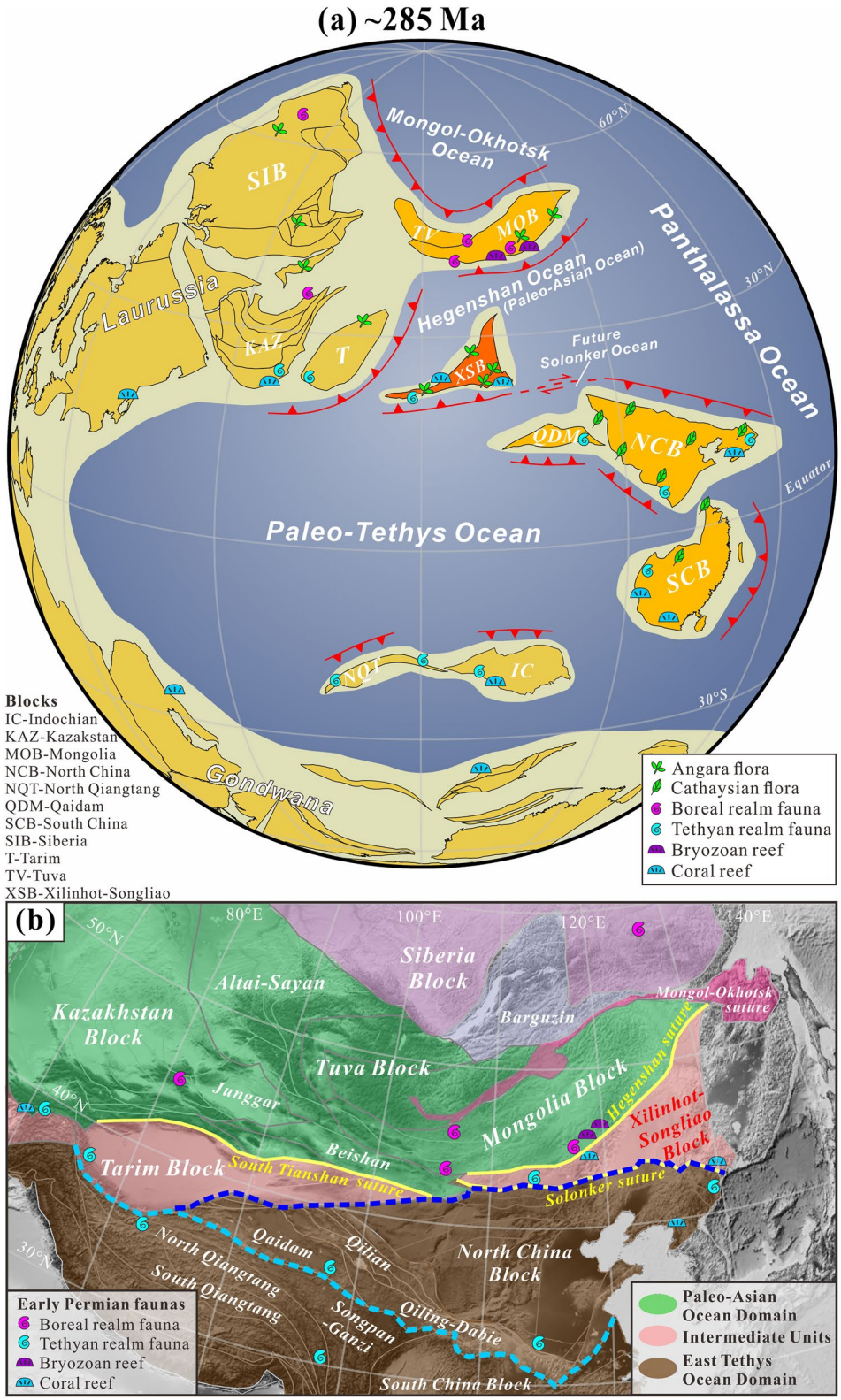


Figure 4.

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record (Shi, 2006; Tian et al., 2016). In addition, detrital zircons from middle Permian sandstones in the MOB and XSB yield different age peaks, indicating that they have different provenance (Eizenhöfer & Zhao, 2018). Therefore, we conclude that the Hegenshan Ocean probably separated the MOB to the north from the XSB to the south at ca. 285 Ma (Figure 4a).

Although the XSB and MOB were separated by the Hegenshan Ocean, the Angara flora were distributed on both blocks (Figure 4a; Sun, 1997). It is possible that the two blocks were connected during the Carboniferous (Eizenhöfer & Zhao, 2018; Ren et al., 2021), when the Angara flora began to develop, and they were later separated by the opening of the Hegenshan Ocean. However, there is still a lack of reliable paleomagnetic constraints on when the ocean basin opened, and this requires further study.

4.2. North–South Range of the East Paleo-Tethys Tectonic Domain

Our new paleogeographic reconstruction redefines the north–south range of the PAO and EPTO domains, and differs from previous models where the North China–Tarim Block separated the PAO and EPTO domains during the late Paleozoic (Figure 4b; e.g., Metcalfe, 2021; Şengör et al., 2018; G. Zhao et al., 2018; Zuza & Yin, 2017). Most early Permian paleogeographic reconstructions have placed the XSB near the northern part of the NCB, with the Solonker Ocean between the two blocks (e.g., Huang et al., 2018; Şengör et al., 2018; Xiao et al., 2018; G. Zhao et al., 2018). However, combining our paleomagnetic data with the distribution of Tethyan fauna suggests that the XSB was located on the northern margin of the EPTO, where oceanic crust was subducted (Figure 4a). Because a plate tectonic domain consists of oceanic crust and related continental crust (Gao, 1993), the northern boundary of the EPTO domain should extend to the southern margin of the XSB (Figure 4b).

The Solonker Ocean, which is generally considered to be the main oceanic basin of the eastern PAO, had not yet formed during the early Permian. Paleomagnetic and geological evidence has suggested that the Solonker Ocean existed as a narrow (~550 km) basin between the XSB and NCB during the Middle Permian (ca. 265 Ma; Y. Liu et al., 2017; Ren et al., 2020; Xiao et al., 2018). We speculate that this limited ocean basin was formed by the westward movement of the NCB (Figure 4a) accompanied by its southward movement (Figure 3b). The XSB and NCB bordered the EPTO and Panthalassa Ocean (PO; Figure 4), respectively, at 285 Ma. The isotopic characteristics of the ophiolites in the Solonker Suture Zone suggest that they belong to both EPTO- and PO-type mantle domains (X. Liu et al., 2021). Therefore, given the potential link between mantle geodynamics and plate tectonics (X. C. Wang et al., 2013), the Solonker Ocean might have belonged to the EPTO–PO tectonic domain rather than the PAO tectonic domain during the Permian. The Hegenshan Ocean between the MOB and XSB may correspond tectonically to the PAO during the Permian (Figure 4a), because the isotopic characteristics of the Hegenshan ophiolite differ from those of the EPTO-type mantle domain (X. Liu et al., 2021). In our new model, the XSB was located between the eastern PAO and the EPTO during the early Permian (Figure 4a) and was the intermediate unit that separated the two tectonic domains (Figure 4b).

5. Conclusions

We obtained new, well-dated (286.1–281.2 Ma) paleomagnetic data from andesites of the Dashizhai Fm. These data meet all seven quality criteria of Meert et al. (2020) and yielded a paleomagnetic pole of 38.4°N/17.6°E (A_{95} = 4.0°) for the XSB at ca. 283.65 Ma. We used our new paleomagnetic data along with available geological and paleontological data to reconstruct the paleogeography of the eastern PAO and EPTO. The XSB and NCB were located at similar paleolatitudes but had different flora. The XSB was located at the northern margin of the EPTO and hosted Tethyan fauna at ca. 285 Ma, suggesting that the northern boundary of the EPTO tectonic domain extended to the southern margin of the XSB. The wide Hegenshan Ocean between the XSB and the MOB was the main PAO basin during the Permian. This new model of the tectonic domains suggests that the XSB was the tectonic unit that separated the eastern PAO and EPTO tectonic domains at ca. 285 Ma.

Figure 4. (a) Schematic paleogeographic reconstruction of the Paleo-Asian Ocean and East Paleo-Tethys Ocean (EPTO) at ~285 Ma. Pangea is shown following the relative Euler rotation parameters given in Domeier and Torsvik (2014). The East Asian blocks are mostly shown according to paleomagnetic constraints (this study and Ren et al., 2020). Distributions of flora and fauna are based on C. Wang et al. (2014), X. Shen et al. (2019) and PBDB (https://paleobiodb.org/). The bryozoon and coral reefs is modified from Tian et al. (2011). (b) New division of the tectonic domains. The light blue dotted line is previous version of the norther boundary of the EPTO domain (e.g., Cawood et al., 2018; Metcalfe, 2021); the dark blue is our new version of the norther boundary of the EPTO tectonic domain.

Data Availability Statement

Data to support this study are presented in Supporting Information S1 and are also available in Zenodo (https://doi.org/10.5281/zenodo.6837170).

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